NEWTON SCHOOL

ANSWER KEY

Linear Equations: Mixed Practice (Equations & Word Problems)

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Part A: Solve Each Equation (1-8)

Problem 1

$\frac{1}{2}(8-4x)=6-2x$

Distributive | Identity

$4-2x=6-2x$
$4=6$ (False)

No Solution

Problem 2

$\frac{1}{3}(15-9a)=7-3a$

Distributive | Identity

$5-3a=7-3a$
$5=7$ (False)

No Solution

Problem 3

$4(3+n)=2(n+9)$

Distributive | Variables Both Sides

$12+4n=2n+18$
$2n=6$

$n=3$

Problem 4

$5(20+s)=6(s+14)$

Distributive | Variables Both Sides

$100+5s=6s+84$
$16=s$

$s=16$

Problem 5

$\frac{8-6y}{2}=y-3$

Fractions | Variables Both Sides

$8-6y=2(y-3)=2y-6$
$8-6y=2y-6$
$14=8y$

$y=\frac{7}{4}$ or 1.75

Problem 6

$\frac{5n-20}{4}=2n+1$

Fractions | Variables Both Sides

$5n-20=4(2n+1)=8n+4$
$-24=3n$

$n=-8$

Problem 7

$2(4y+3)-y=3(y-1)+5$

Distributive | Combine Like Terms

$8y+6-y=3y-3+5$
$7y+6=3y+2$
$4y=-4$

$y=-1$

Problem 8

$5(2y-3)+8=3y+7$

Distributive | Variables Both Sides

$10y-15+8=3y+7$
$10y-7=3y+7$
$7y=14$

$y=2$

Part B: Number Word Problems (9-14)

Problem 9

Find a number that is 72 greater than its opposite.

Opposite

Let x = the number

$x = -x + 72$

$2x = 72$

Equation: $x = -x + 72$; Answer: $x = 36$

Problem 10

Find a number that is 54 less than its opposite.

Opposite

Let x = the number

$x = -x - 54$

$2x = -54$

Equation: $x = -x - 54$; Answer: $x = -27$

Problem 11

Find a number whose product with 7 is the same as its sum with 42.

Product Sum

Let x = the number

$7x = x + 42$

$6x = 42$

Equation: $7x = x + 42$; Answer: $x = 7$

Problem 12

Find a number that is 84 greater than four times its opposite.

Opposite Multiple

Let x = the number

$x = 4(-x) + 84$

$x = -4x + 84$

$5x = 84$

Equation: $x = -4x + 84$; Answer: $x = \frac{84}{5} = 16.8$

Problem 13

Five times a number, decreased by 12, is the same as three times the number, increased by 8.

Variables Both Sides

Let x = the number

$5x - 12 = 3x + 8$

$2x = 20$

Equation: $5x - 12 = 3x + 8$; Answer: $x = 10$

Problem 14

The greater of two consecutive odd integers is 14 more than twice the smaller.

Consecutive Odd

Let x = smaller odd integer

$x + 2$ = greater odd integer

$x + 2 = 2x + 14$

$-12 = x$

Equation: $x + 2 = 2x + 14$; Answer: -12 and -10

Part C: Applied Word Problems (15-20)

Problem 15

Maria scored twice as many points as Alex, while Ben scored 8 more than Alex. Maria and Ben scored the same.

Comparison

Let x = Alex's points

Maria = $2x$, Ben = $x + 8$

$2x = x + 8 \rightarrow x = 8$

Alex: 8, Maria: 16, Ben: 16

Problem 16

Tom has 8 identical weights. 7 weights = 1 weight + 150g.

Balance

Let w = mass of each weight

$7w = w + 150$

$6w = 150$

Equation: $7w = w + 150$; Answer: 25 g

Problem 17

Sum of two numbers is 24. Four times one is 6 less than six times the other.

Sum System

Let x = one number, $24 - x$ = other

$4x = 6(24 - x) - 6$

$4x = 144 - 6x - 6 = 138 - 6x$

$10x = 138$, $x = 13.8$

Equation: $4x = 6(24 - x) - 6$; Numbers: 13.8 and 10.2

Problem 18

Difference of two integers is 12. Four times smaller is 8 more than twice larger.

Difference

Let x = smaller, $x + 12$ = larger

$4x = 2(x + 12) + 8$

$4x = 2x + 24 + 8 = 2x + 32$

$2x = 32$, $x = 16$

Equation: $4x = 2(x + 12) + 8$; Integers: 16 and 28

Problem 19

Triangle sides are consecutive odd integers. Longest side is 20 less than perimeter.

Consecutive Odd Perimeter

Sides: x, $x + 2$, $x + 4$

Perimeter = $3x + 6$

$x + 4 = (3x + 6) - 20$

$x + 4 = 3x - 14$, $18 = 2x$, $x = 9$

Equation: $x + 4 = 3x + 6 - 20$; Longest side: 13

Problem 20

Ryan: 20+2/week. Emma: 32+0.50/week. When equal?

Linear Model

Ryan: $20 + 2w$

Emma: $32 + 0.5w$

$20 + 2w = 32 + 0.5w$

$1.5w = 12$, $w = 8$

Equation: $20 + 2w = 32 + 0.5w$; Answer: 8 weeks

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